Image Dehazing using Adaptive Reconstruction

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1 Introduction

Outdoor images captured in real-world environments often suffer from degradation caused by atmospheric conditions such as haze, fog, dust, and smoke. These particles scatter light before it reaches the camera sensor, which significantly reduces the quality and clarity of captured images.

This degradation results in poor visibility and color distortion. As a consequence, many computer vision systems that rely on clear images perform poorly when haze is present.

1.1 Impact of Haze on Images

Haze causes two major physical effects in images:

- Attenuation of scene radiance due to scattering
- Addition of atmospheric light along the line of sight

These effects lead to:

- Reduced image contrast
- Loss of scene details
- Blurred edges
- Color distortion

1.2 Applications Affected by Haze

Image degradation caused by haze can negatively affect several important applications including:

- Autonomous driving
- Surveillance systems
- Remote sensing
- Medical imaging
- Object detection systems

Therefore, restoring clear images from hazy observations is an important research problem.

1.3 Motivation of the Work

Traditional image enhancement methods such as histogram equalization improve contrast but do not properly model the physical formation of haze.

Therefore, modern dehazing approaches rely on the atmospheric scattering model to estimate the true scene radiance.

This project proposes an uncertainty-aware framework that combines:

- transmission estimation
- regularized reconstruction
- multi-sample restoration
- frequency-domain fusion

This approach improves the stability and robustness of dehazing under varying haze conditions.

2 Problem Statement

Recovering the clean scene radiance from a hazy image is a challenging task because the haze formation process depends on unknown scene depth and atmospheric parameters.

2.1 Ill-posed Nature of Dehazing

The problem is considered ill-posed because a single hazy image contains multiple unknown variables such as:

- atmospheric light
- transmission map
- scene depth

Without additional information, directly recovering the original scene becomes difficult.

2.2 Limitations of Direct Inversion

Direct inversion of the haze model often produces unstable results when the transmission value is very small.

This can lead to:

- noise amplification
- unrealistic colors
- artifacts in the recovered image

2.3 Need for Robust Reconstruction

To address these challenges, a robust reconstruction method must:

- estimate transmission reliably
- prevent noise amplification
- handle different haze levels
- preserve image details

3 Objectives

The main objectives of this project are:

1. Model haze formation using the atmospheric scattering model.
2. Estimate atmospheric light from hazy images.
3. Estimate the transmission map.
4. Recover the clean image using regularized pseudo-inverse reconstruction.
5. Generate multiple reconstructions using uncertainty sampling.
6. Fuse reconstructed images using frequency-domain fusion.
7. Evaluate haze removal using Frobenius norm analysis.

4 Atmospheric Scattering Model

4.1 Image Formation under Haze

The formation of a hazy image can be described using the atmospheric scattering model.

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{1}$$

where

- $I(x)$ represents the observed hazy image
- $J(x)$ represents the true scene radiance
- $t(x)$ represents the transmission map
- A represents the atmospheric light

4.2 Transmission Map

The transmission map indicates how much light from the scene reaches the camera.

It depends on the scene depth and the scattering coefficient.

$$t(x) = e^{-\beta d(x)} \tag{2}$$

where

- $d(x)$ represents scene depth
- β represents the scattering coefficient

4.3 Scene Radiance Recovery

Rearranging the atmospheric scattering equation allows recovery of the clean image.

$$J(x) = \frac{I(x) - A(1 - t(x))}{t(x)} \tag{3}$$

However, when transmission values are small, the division may cause instability.

5 Methodology

5.1 Atmospheric Light Estimation

Atmospheric light represents the intensity of scattered light in the atmosphere.

It is estimated by analyzing the brightest pixels in the hazy image.

Steps involved:

1. Convert the RGB image into grayscale.
2. Sort the pixel intensities.
3. Select the top 0.1% brightest pixels.
4. Compute the average intensity of these pixels.

$$A = mean(top\ bright\ pixels) \tag{4}$$

5.2 Transmission Map Estimation

Transmission estimation is performed using a local averaging filter applied to the grayscale image.

$$t(x) = 1 - blur(I_{gray}) \tag{5}$$

To avoid extreme transmission values, the map is clamped.

$$t(x) = \min(\max(t(x), 0.3), 0.9) \tag{6}$$

This ensures stable image reconstruction.

5.3 Haze Strength Estimation

The global haze level in the image is estimated using the mean value of the transmission map.

$$H = mean(1 - t(x)) \tag{7}$$

If

- $H > 0.6$ the image is considered thick haze
- $H \leq 0.6$ the image is considered thin haze

5.4 Regularized Pseudo-Inverse Reconstruction

Direct inversion of the haze model can produce unstable results when transmission is very small.

To overcome this problem, a regularized pseudo-inverse formulation is used.

$$H_{pinv} = \frac{t}{t^2 + \lambda} \tag{8}$$

where λ is the regularization parameter.

The reconstructed image is computed as:

$$J = H_{pinv}(I - A) + A \tag{9}$$

5.5 Uncertainty Sampling

Since parameter estimation may contain uncertainty, multiple reconstructed images are generated using different parameter settings.

$$J_1, J_2, J_3, J_4 \tag{10}$$

Each reconstruction uses different blur sizes and regularization parameters.

5.6 Frequency Domain Fusion

The reconstructed images are combined using frequency-domain averaging.

The 2D Fourier transform of an image is defined as:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi\left(\frac{ux}{M} + \frac{vy}{N}\right)} \tag{11}$$

The fused frequency spectrum is obtained by averaging:

$$F_{fused} = \frac{1}{N} \sum_{k=1}^{N} F_k \tag{12}$$

The final image is recovered using the inverse Fourier transform.

$$f(x,y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u,v) e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)} \tag{13}$$

5.7 Patch-wise Haze Modeling

In practical scenarios haze is rarely uniform across the entire scene. Instead, haze often appears in localized regions depending on atmospheric conditions and scene depth. To evaluate the robustness of the proposed dehazing framework under spatially varying haze conditions, synthetic patch-wise haze is generated on a clean image.

Let $J(x)$ denote the clean image and A denote the atmospheric light. The transmission map is initially defined as

$$t(x) = 1 \tag{14}$$

which represents a haze-free region.

Two rectangular regions P_1 and P_2 are selected in the image domain to simulate localized haze patches. Within each patch, the transmission varies radially from the center of the patch to the boundary.

Let (x_c, y_c) denote the center of the patch. The radial distance of a pixel (x, y) from the center is defined as

$$d(x,y) = \sqrt{(x-x_c)^2 + (y-y_c)^2} \tag{15}$$

To normalize the distance within the patch, the following expression is used

$$d_n(x,y) = \frac{d(x,y)}{\max(d(x,y))} \tag{16}$$

The transmission inside the patch is defined as a radial function

$$t_p(x,y) = t_{min} + \alpha d_n(x,y) \tag{17}$$

where

- t_{min} represents the minimum transmission value (dense haze region)
- α controls the strength of the transmission gradient

In the implemented experiment the parameters are selected as

$$t_p(x, y) = 0.4 + 0.3d_n(x, y) \tag{18}$$

which produces transmission values between 0.4 and 0.7.

The global transmission map is therefore defined as

$$t(x, y) = \begin{cases} t_p(x, y) & (x, y) \in P_1 \cup P_2 \\ 1 & \text{otherwise} \end{cases} \tag{19}$$

To avoid sharp discontinuities at patch boundaries, the transmission map is smoothed using a Gaussian filter

$$t_s(x, y) = G_\sigma * t(x, y) \tag{20}$$

where G_σ denotes the Gaussian kernel and $*$ represents convolution.

Using this transmission map, the synthetic hazy image is generated according to the atmospheric scattering model

$$I(x, y) = J(x, y)t_s(x, y) + A(1 - t_s(x, y)) \tag{21}$$

This produces a hazy image containing spatially varying haze regions.

5.8 Patch-Selective Dehazing

After performing global dehazing using the proposed reconstruction framework, only the regions affected by haze should be corrected while preserving already clear regions.

To identify haze regions, a binary haze mask is constructed based on the transmission map

$$M(x, y) = \begin{cases} 1 & t(x, y) < \tau \\ 0 & t(x, y) \geq \tau \end{cases} \tag{22}$$

where τ is a threshold value. In this implementation $\tau = 0.98$.

To ensure smooth transitions between corrected and uncorrected regions, the mask is further smoothed using Gaussian filtering

$$M_s(x, y) = G_\sigma * M(x, y) \tag{23}$$

Let $J_g(x, y)$ denote the globally dehazed image obtained from the reconstruction stage. The final patch-selective reconstruction is computed as

$$J_{final}(x, y) = M_s(x, y)J_g(x, y) + (1 - M_s(x, y))I(x, y) \tag{24}$$

where

- $J_g(x, y)$ represents the globally reconstructed image
- $I(x, y)$ represents the original hazy image
- $M_s(x, y)$ controls the contribution of the reconstructed image

This formulation ensures that dehazing is applied primarily to regions affected by haze, while preserving the original appearance of already clear areas of the image.

5.9 Contrast Enhancement and RGB color correction

For images with strong haze, adaptive histogram equalization is applied to enhance local contrast and reveal hidden details. After dehazing, color imbalance may occur.

Channel-wise scaling is applied to restore natural color appearance.

$$scale = \frac{mean(I)}{mean(J)} \tag{25}$$

The corrected image is:

$$J_{corrected}(c) = J(c) \times scale(c) \tag{26}$$

6 Evaluation using Frobenius Norm

The Frobenius norm is used to measure the energy of an image.

$$||I||_F = \sqrt{\sum I^2} \tag{27}$$

$$||J||_F = \sqrt{\sum J^2} \tag{28}$$

The amount of haze removed is calculated as:

$$||I - J||_F \tag{29}$$

The percentage haze reduction is given by:

$$Haze\ Reduction = \frac{||I - J||_F}{||I||_F} \times 100 \tag{30}$$

7 Results and Discussion

The proposed image dehazing framework was evaluated on several hazy images with different haze intensities. The experimental results demonstrate that the proposed approach effectively improves image visibility, restores scene details, and enhances overall image quality.

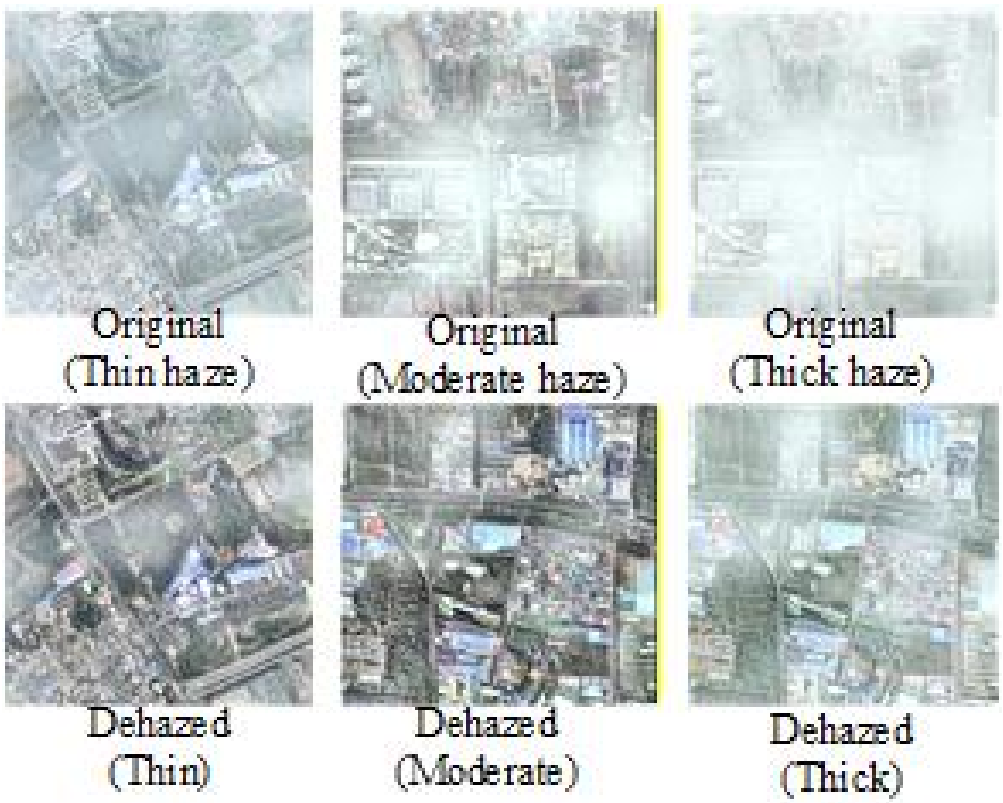

Figure 1: Dehazing result obtained using the proposed method showing with different levels of haziness.

One of the main advantages of the proposed framework is the use of multiple reconstruction samples. Instead of relying on a single estimate of the transmission map, the algorithm

generates several reconstructed images using different blur sizes and regularization parameters. This multi-sample strategy helps capture the uncertainty present in transmission estimation. By considering multiple possible reconstructions, the method avoids over-reliance on a single parameter configuration and improves the robustness of the final output.

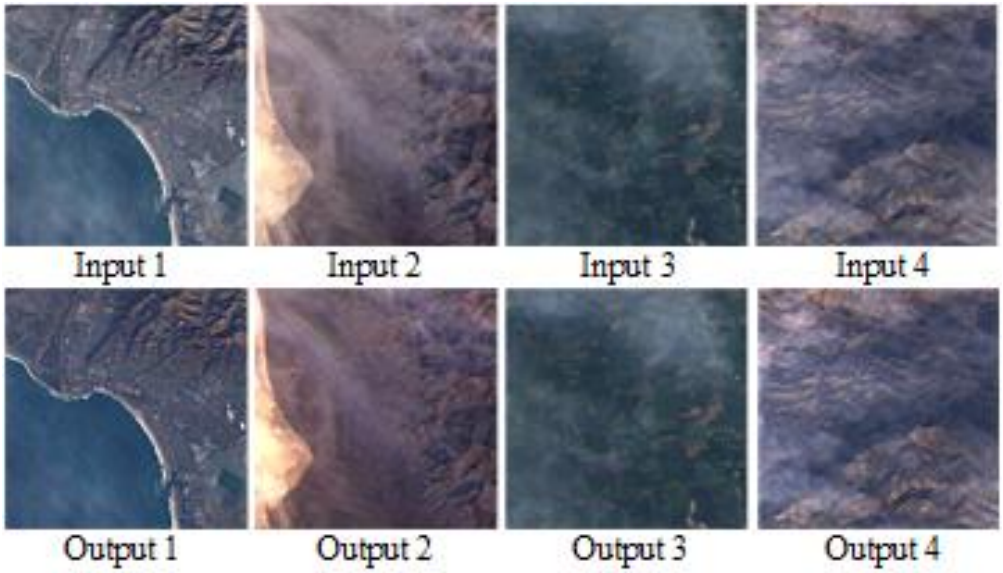

Figure 2: Dehazing result obtained using the proposed method showing in different regions.

The reconstructed images are combined using frequency-domain fusion. In this step, each reconstructed image is transformed into the frequency domain using the Fourier transform. Averaging the frequency spectra allows the algorithm to combine useful information from each reconstruction. Low-frequency components contribute to global brightness and haze correction, while high-frequency components preserve fine details such as edges and textures. As a result, the fused image achieves a balanced reconstruction with improved contrast and preserved structural information.

Figure 3: Dehazing result obtained using the proposed method with patchwise haziness.

Visual inspection of the output images shows several improvements compared to the original hazy images. The restored images exhibit significantly improved visibility, allowing distant objects and scene structures to become more distinguishable. Edges and textures that were previously obscured by haze become clearer after the dehazing process. This improvement is particularly noticeable in regions with strong haze accumulation, where traditional enhancement techniques often fail.

Another important observation is the improvement in color consistency. Hazy images typically suffer from washed-out colors due to atmospheric light scattering. The proposed method addresses this issue through RGB color correction, which rescales the color channels to maintain a natural appearance. As a result, the dehazed images show more balanced color distribution and better visual realism.

```
===== FROBENIUS NORM ANALYSIS =====
Input (original target): 438.1119
Input (synthetic haze): 474.7369
Output        : 444.4685
Haze Removed : 67.9017
Haze Reduction %  : 14.303
```

Figure 4: Frobenius comparison for patchwise haziness.

```
FROBENIUS NORM ANALYSIS
Input          : 672.7853
Output         : 564.9371
Haze Removed   : 149.7197
Haze Reduction %  : 22.2537
```

Figure 5: Frobenius comparison for Thin haziness.

```
FROBENIUS NORM ANALYSIS
Input          : 799.4276
Output         : 717.0184
Haze Removed   : 112.4214
Haze Reduction %  : 14.0627
```

Figure 6: Frobenius comparison for Thick haziness.

```
FROBENIUS NORM ANALYSIS
Input          : 680.6378
Output         : 573.6836
Haze Removed   : 145.4408
Haze Reduction %  : 21.3683
```

Figure 7: Frobenius comparison for Moderate haziness.

Contrast enhancement further improves image clarity, especially in cases of thick haze. Adaptive histogram equalization increases local contrast, making subtle details more visible. This step helps reveal fine structures that are otherwise hidden in heavily degraded regions.

The effectiveness of haze removal can also be quantified using the Frobenius norm analysis described earlier. By comparing the energy of the input hazy image and the reconstructed image, it is possible to estimate the amount of haze removed by the algorithm. The reduction in Frobenius norm difference indicates that a significant portion of haze-related degradation has been successfully removed.

Overall, the experimental results confirm that the proposed uncertainty-aware reconstruction framework provides stable and visually pleasing dehazing results. The combination of transmission estimation, regularized pseudo-inverse reconstruction, multi-sample restoration, and frequency-domain fusion contributes to improved robustness and better visual quality compared to simple single-step reconstruction approaches.

8 Conclusion

This project presented an uncertainty-aware image dehazing framework designed to restore clear images from hazy observations. The proposed method is based on the atmospheric scattering model and incorporates several processing stages to improve reconstruction stability and image quality.

The first stage of the framework estimates atmospheric light and transmission maps from the input hazy image. These parameters describe how haze affects the observed scene and

form the basis for reconstructing the clean image. However, direct inversion of the haze model can produce unstable results when transmission values are small. To address this issue, a regularized pseudo-inverse reconstruction method was introduced. The regularization parameter helps prevent noise amplification and stabilizes the reconstruction process.

To further improve robustness, the method generates multiple reconstructed images using different parameter settings. This multi-sample reconstruction strategy captures the uncertainty inherent in transmission estimation and produces several possible clean images. Instead of selecting a single reconstruction, the algorithm combines these images using frequency-domain fusion. By averaging their Fourier representations, the method integrates complementary information from multiple reconstructions while preserving important image structures.

Additional processing steps such as contrast enhancement and RGB color correction further improve the visual quality of the final output. These steps help restore natural colors and enhance local contrast, making the dehazed images visually clearer and more realistic.

Experimental results demonstrate that the proposed approach effectively reduces haze, improves visibility, and restores important scene details. The combination of uncertainty sampling and frequency fusion produces stable reconstructions that maintain both global brightness and fine textures. The Frobenius norm evaluation also confirms that the proposed framework successfully reduces haze-related degradation.

In summary, the proposed method provides a reliable and effective solution for single-image dehazing. The framework balances physical modeling and signal processing techniques to achieve high-quality image restoration.

Overall, the proposed framework provides a strong foundation for further research in robust image dehazing and atmospheric image restoration.